

CLWE-PRUNE: CROSS-LINGUAL WORST-GROUP EXCESS FOR MULTILINGUAL TEXT-TO-SPEECH COMPRESSION

Cuicui Zhu¹, Jiawei Zhan¹, Xiaoyong Guo¹, Hao Huang^{1,2,3,†}, Kai Wang^{1,2,3,†}, Yongchao Wang⁴

¹*School of Computer Science and Technology, Xinjiang University, Urumqi, China*

²*Joint International Research Laboratory of Silk Road
Multilingual Cognitive Computing, Urumqi, China*

³*Xinjiang Key Laboratory of Multilingual Information Technology, Urumqi, China*

⁴*Department of Science and Technology Intelligence, Marketing Service Center of
State Grid Xinjiang Electric Power Co., Ltd., China*

zhu16198@gmail.com, {wangkai, huanghao}@xju.edu.cn, im_guoxy@163.com

ABSTRACT

English-only layer selection can overlook degradation in other languages when compressing multilingual text-to-speech (TTS) models. We propose Cross-Lingual Worst-Group Excess Pruning (CLWE-Prune), which scores each large language model (LLM) block by the largest increase in recognition error caused by its removal across English, Chinese, Japanese, and Korean. We prune the lowest-scoring blocks and fine-tune the pruned models on student-generated speech-token sequences screened for text accuracy using offline automatic speech recognition (ASR). On CosyVoice3, CLWE-selected P_k models achieve lower four-language average Whisper error (Macro ER) than EN-WER P_k at all four matched pruning depths. With 400 verified sequences and 15 updates, recovery reduces Whisper Macro ER by 7.61–10.64% relative to Base at P_{12} , P_{15} , and P_{16} , and lowers SenseVoice Macro ER at all three depths. With 800 sequences and 40 updates, recovered P_{15} retains nine of 24 LLM blocks and has 26.03% fewer parameters than Dense-24. Its observed Whisper and SenseVoice Macro ERs are 7.68% and 8.73%, respectively, compared with 7.86% and 8.77% for Dense-24. In a separate inference benchmark, a P_{15} checkpoint reduces the real-time factor (RTF) from 0.633 to 0.375 and first-packet latency from 2.600 to 1.572 s relative to Dense-24.

Index Terms— Multilingual TTS, Model Compression, Structural Pruning, Cross-Lingual Layer Selection

1. INTRODUCTION

Multilingual text-to-speech (TTS) systems increasingly use a shared Transformer backbone to synthesize speech across languages [1, 2]. Neural TTS has improved acoustic modeling and waveform generation [3]. Prompt-conditioned systems can synthesize speech for unseen speakers across languages [4, 5], while speech-token models cast generation as autoregressive sequence prediction [6, 7]. This shared backbone, however, makes streaming inference costly: each autoregressive decoding step traverses all Transformer blocks sequentially. Removing entire blocks can shorten this path. However, a block that appears dispensable for English may still be crucial for preserving linguistic content in languages such as Chinese, Japanese, and Korean. Aggregate error alone can obscure this risk: the dense

model has different baseline error rates across languages, while English is evaluated using word error rate (WER) and the other languages using character error rate (CER). We therefore focus on the *additional* error caused by pruning within each language, particularly the largest such increase across languages. Acoustic quality scores may remain stable even as errors in linguistic content increase.

Structural pruning removes dimensions or layers [8, 9, 10]. SPADE applies WER-guided layer selection and distillation to LLM-based TTS [11], but its English-only ranking cannot detect degradation in Chinese, Japanese, or Korean. Averaging error rates across languages can mask a substantial increase for an individual language, motivating worst-group evaluation for model selection [12]. Pruning can also alter which languages are adversely affected, as shown for multilingual text LLMs under mismatched calibration [13]. ASR-screened self-distillation has been used to reduce synthesis failures in neural-codec TTS [14]. To our knowledge, however, cross-lingual worst-group block selection combined with ASR-screened student-trajectory recovery has not been studied for pruned multilingual TTS.

Pruning also changes the student’s generation behavior. Standard distillation trains compact models using supervision from a teacher [15, 16]. Sequences produced by a dense teacher or extracted from recordings differ from those generated by the pruned student. On-policy distillation addresses this mismatch by learning from the student’s own generated sequences [17], and self-training work shows that generated supervision must be screened before use [18]. Using a student-generated sequence as a training target can reinforce pronunciation errors in the decoded speech. For TTS, offline ASR can compare the decoded speech with the requested text and discard trajectories that fail this content check. Screening is performed only during training and adds no cost to the deployed synthesizer.

We propose CLWE-Prune, which ranks layers by the worst-group excess error across four language groups. It then recovers each pruned model using ASR-screened, student-generated speech-token trajectories while keeping the acoustic decoder fixed. Our contributions are:

- We propose a cross-lingual layer-selection criterion based on the largest increase in recognition error relative to the dense model, improving multilingual content preservation over English-only ranking.

[†]Corresponding authors.

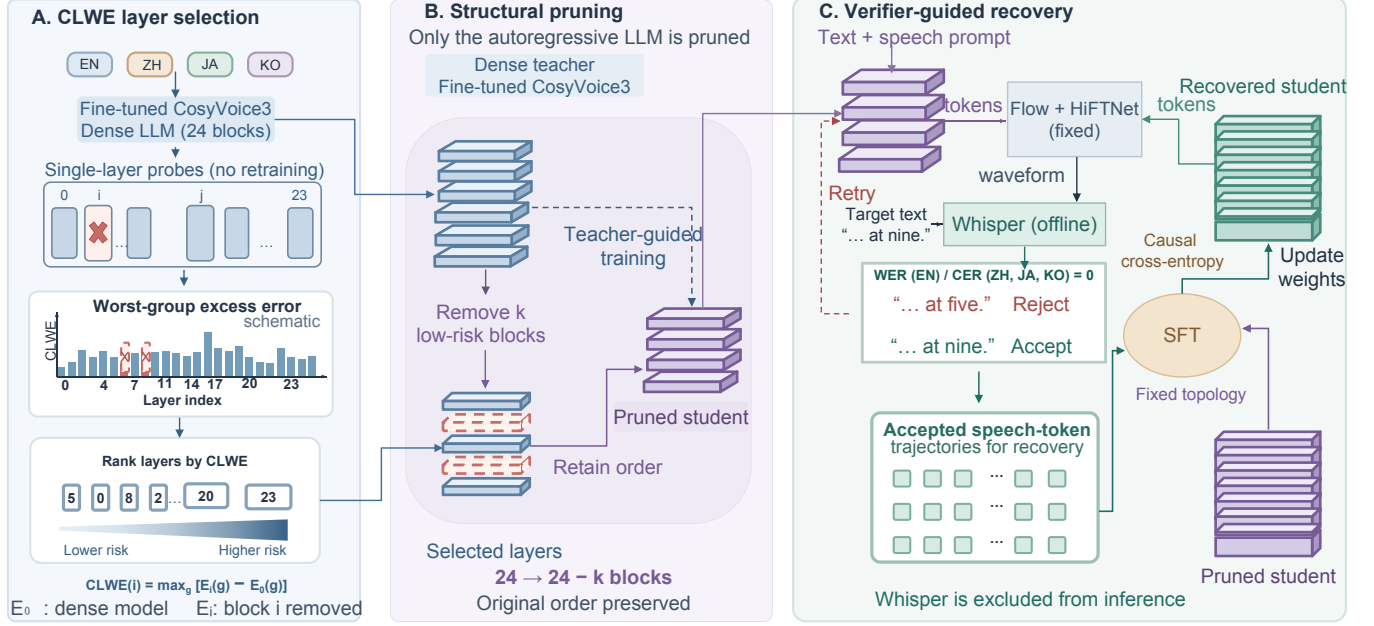


Fig. 1. Overview of CLWE-Prune. (A) Single-layer removal estimates cross-lingual worst-group excess risk. (B) The lowest-scoring blocks are removed, and the remaining model undergoes continuation training to obtain the Base checkpoint, with the acoustic decoder fixed. (C) The pruned student is fine-tuned on ASR-screened speech-token trajectories with its pruning topology fixed.

- We introduce verifier-guided recovery using ASR-screened, student-generated trajectories. This reduces recognition errors without changing the pruned architecture or acoustic decoder.
- Our recovered nine-layer model removes 26.03% of the dense model’s parameters while achieving comparable recognition accuracy and a $1.69\times$ speedup in streaming inference, measured by RTF.

2. METHOD

Figure 1 summarizes three stages: single-layer risk estimation, physical pruning, and verifier-guided recovery. We rank CosyVoice3’s 24 autoregressive LLM blocks by their cross-lingual worst-group excess error and remove the lowest-risk blocks. Retained blocks keep their original order and Dense-24 weights. Under the matched-selection protocol, the pruned LLM undergoes continuation training guided by a frozen Dense-24 teacher; the checkpoint with the lowest validation loss becomes the Base model. Recovery fine-tunes Base on ASR-screened, student-generated speech-token trajectories. The teacher used to construct Base is distinct from the Dense-Teacher recovery control, which supplies recovery targets. The flow-matching acoustic model and HiFTNet vocoder remain unchanged throughout.

2.1. CLWE-Based Layer Selection

As shown in Fig. 1(A), we estimate the risk of removing a single block using a multilingual layer-selection dataset. We partition the samples into language groups $\mathcal{G} = \{g_{\text{EN}}, g_{\text{ZH}}, g_{\text{JA}}, g_{\text{KO}}\}$.

To probe block i , we remove only that block from Dense-24 and evaluate the remaining model without retraining. All single-block probes use the same inference and scoring procedure. Let $E_i(g)$ and $E_0(g)$ denote the recognition errors of the probe and Dense-24, respectively, on language group g .

Following SPADE’s WER-based layer-importance approach [11], the English-only baseline ranks block i by $R_i^{\text{EN}} = E_i(g_{\text{EN}})$. Subtracting the constant $E_0(g_{\text{EN}})$ from every English score would leave this ranking unchanged. This baseline compares layer-selection criteria; it does not reproduce the full SPADE training pipeline.

CLWE instead measures the additional error caused by removing each block within each language:

$$\begin{aligned} \Delta_i(g) &= E_i(g) - E_0(g), \\ R_i^{\text{CLWE}} &= \max_{g \in \mathcal{G}} \Delta_i(g). \end{aligned} \quad (1)$$

Subtracting $E_0(g)$ accounts for each language’s Dense-24 baseline, while taking the maximum sets the block’s risk score to the largest observed error increase. This subtraction does not make English WER statistically equivalent to the other languages’ CER. A lower R_i^{CLWE} indicates a safer block to remove.

We rank blocks in ascending order of risk, breaking ties by their original indices. For a pruning budget of k , we physically remove the first k blocks in this ranking, as shown in Fig. 1(B). The ranking is computed once and reused across pruning depths, yielding nested removal sets that preserve the retained blocks’ order. Single-block probes do not capture interactions among jointly removed blocks; consequently, their ranking is a selection heuristic, not a bound on the final pruned model’s error.

2.2. Verifier-Guided Student-Trajectory Recovery

As shown in Fig. 1(C), each pruned Base model generates speech-token trajectories $\mathbf{z} = (z_1, \dots, z_T)$ conditioned on text x and prompt c . Generation uses temperature 1.0, $\text{top-p}=0.8$, and $\text{top-k}=25$. The fixed flow-matching acoustic model and HiFTNet vocoder decode the tokens into speech. After language-specific text normalization, Whisper evaluates content using WER for English and CER for Chinese, Japanese, and Korean.

Zero-error candidates are prioritized. When an input has no acceptable candidate, we generate additional candidates or draw from a shared reserve text pool. Among candidates passing the content check, speaker similarity, UTMOS, and duration ratio guide selection; these quality measures do not replace the ASR criterion. Candidate generation and screening are performed offline.

Each pruned Base model first generates 400 speech-token trajectories, from which we select recovery targets using the verifier. We fine-tune the student on these targets for 15 updates to obtain SFT400/15. This checkpoint then generates a second set of 400 verifier-screened targets.

For SFT800/40, we reinitialize the student from its original Base checkpoint and fine-tune it on both target sets for 40 updates. SFT400/15 supplies the second target set, not the initialization. In the P_{12} Base-generated control, the original Base model generates both target sets and initializes the final model.

Let $\mathcal{D}_{\text{rec}}$ contain the accepted trajectories. For sequence j in a minibatch $\mathcal{B}$, set $\tilde{z}_{j,t} = z_{j,t}$ for $t \leq T_j$ and $\tilde{z}_{j,T_j+1} = \text{EOS}$. With $h_{j,t} = (x_j, c_j, \tilde{z}_{j,<t})$, we optimize

$$\mathcal{L}_{\text{rec}} = -\frac{1}{N_{\mathcal{B}}} \sum_{j,t} \log p_{\theta_P}(\tilde{z}_{j,t} | h_{j,t}). \quad (2)$$

Here $\sum_{j,t}$ ranges over $j \in \mathcal{B}$ and $t = 1, \dots, T_j + 1$, while $N_{\mathcal{B}} = \sum_{j \in \mathcal{B}} (T_j + 1)$ counts valid targets, excluding masked text and padding positions. The parameters θ_P belong to the pruned LLM. Recovery uses a learning rate of 3×10^{-6} and mixed precision. Teacher forcing trains the LLM to predict the accepted sequences; Whisper provides no training gradients.

2.3. Experimental Setup

Data. The calibration set contains 400 utterances per language-corpus group from LibriTTS (English) [19], AISHELL-3 (Chinese) [20], and Emilia (Japanese and Korean) [21]. The `strict1200` test set contains 300 utterances per language. All models use the same test texts and prompts but generate separate waveforms. Recovery uses 100 and 200 targets per language for SFT400/15 and SFT800/40, respectively.

Models and training configurations. Dense-24 is the unpruned 24-block model. Removing k LLM blocks leaves $24 - k$ blocks. Multilingual configurations selected by CLWE at this depth are labeled P_k , and the English-only pruning baseline is labeled EN-WER P_k . Table 1 compares these configurations at $k \in \{6, 8, 10, 12\}$. Table 2 extends the multilingual sweep to P_{13} – P_{17} and reports a separate inference benchmark for Dense-24, P_{12} , P_{15} , and P_{16} .

Table 3 compares recovery from P_{12} , P_{15} , and P_{16} Base checkpoints, which are distinct from those used in the sweep in Table 2. Verifier-SFT is evaluated at both 400/15 and 800/40 for all three depths. At P_{12} , additional controls include Base-Generated-SFT800/40, Selected-Original400/15 using tokens extracted from original recordings, and Random-Candidate400/15 using unverified student-generated targets with separately assigned same-language prompts. Dense-Teacher-SFT400/15 uses verified dense-model targets at all three depths. Dense-24 Teacher is the unpruned reference. **Evaluation metrics.** Content accuracy is evaluated using Whisper large-v3 [22] and SenseVoiceSmall [23]. WER is used for English, while CER is used for Chinese, Japanese, and Korean. We report macro-averaged error rate (Macro ER), computed as the mean of the four language-level error rates. For each fixed checkpoint, Table 3 reports results averaged over three decoding runs with seeds 1986, 1987, and 1988. For a model evaluated alongside Dense-24, the worst-group excess on the test set is $\max_g [E_{\text{model}}(g) - E_{\text{Dense}}(g)]$.

We report UTMOS quality predictions [24] and mel-cepstral distortion (MCD) in the recovery comparison. The high-compression sweep also reports the cosine similarity of CAM++ speaker embeddings (SECS) [25]. MCD is computed from 24th-order mel cepstra (excluding c_0) extracted using WORLD at 24 kHz and aligned with fastDTW. Higher UTMOS and SECS and lower MCD indicate better performance. UTMOS is also used to rank candidates. These automatic scores do not establish perceived naturalness. A separate architecture benchmark measures streaming real-time factor (RTF), first-packet latency, and peak allocated CUDA memory.

Table 1. English-WER versus multilingual CLWE layer selection. Bold indicates values for which P_k outperforms EN-WER P_k at the same depth.

Model	Macro↓	EN/ZH↓	JA/KO↓	UTMOS↑/MCD↓
EN-WER P6	7.92	2.16/12.17	12.12/5.26	3.47/8.32
P6	7.67	2.75/ 11.67	11.92/4.34	3.48/8.28
EN-WER P8	7.91	2.23/12.67	11.93/4.81	3.47/8.32
P8	7.65	2.04/12.01	12.16/ 4.37	3.48/8.26
EN-WER P10	8.47	2.81/11.97	13.85/5.27	3.48/8.27
P10	7.78	2.53/12.28	11.45/4.88	3.47/8.29
EN-WER P12	8.97	2.64/12.10	16.25/4.90	3.49/8.31
P12	8.09	2.42/12.38	12.51/5.04	3.48/8.32

2.4. Results and Analysis

Table 2. High-compression quality sweep and separate architecture inference benchmark. The Layers column reports the number of retained blocks. Macro ER refers to Whisper Macro ER.

Model	Layers	Macro ER↓	UTMOS↑	SECS↑	MCD↓
P13	11	8.36	3.48	0.80	8.31
P14	10	8.65	3.48	0.80	8.36
P15	9	8.75	3.48	0.80	8.35
P16	8	11.31	3.47	0.80	8.40
P17	7	13.17	3.48	0.80	8.46

Model	Params (M)↓	RTF↓	First pkt. (s)↓	Peak CUDA (MiB)↓
Dense-24	859.19	0.633	2.600	3382
P12	680.24	0.434	1.779	2697
P15	635.50	0.375	1.572	2530
P16	620.59	0.377	1.535	2468

At matched pruning depths (Table 1), the multilingual P_k models reduce Whisper Macro ER relative to EN-WER by 0.25, 0.26, 0.69, and 0.88 percentage points at P_6 , P_8 , P_{10} , and P_{12} , respectively. The maximum reported error across languages also decreases by 0.25, 0.51, 1.57, and 3.74 percentage points, respectively. This descriptive maximum combines English WER with CER for the other languages. At P_{12} , Japanese CER falls from 16.25% to 12.51%, illustrating the non-English degradation overlooked by English-only ranking. At P_{10} , however, Chinese CER increases by 0.31 percentage points, so the gains are not uniform. Mean-Excess and Max-Raw checkpoints were not evaluated, so the effects of taking the maximum and subtracting the dense-model baseline are not isolated. The matched comparison ends at P_{12} .

Table 2 reports the available multilingual P_k configurations rather than a controlled ablation of pruning depth alone. Whisper

Table 3. Recovery performance on `strict1200`. W and SV denote Whisper large-v3 and SenseVoiceSmall. Bold indicates the best value at each pruning depth; Dense-24 Teacher is the unpruned reference.

Recovery Model	W-Macro↓	SV-Macro↓	EN W/SV	ZH W/SV	JA W/SV	KO W/SV	UTMOS↑	MCD↓
P12 (Base)	8.15	9.23	2.64/4.11	12.36/10.86	12.36/14.19	5.23/7.76	3.4884	8.3143
P12 + Verifier-SFT400/15	7.53	8.40	2.61/3.75	11.79/10.56	11.16/12.41	4.58/6.87	3.5241	8.2449
P12 + Verifier-SFT800/40	7.24	8.32	2.36/3.68	11.51/10.45	10.76/12.39	4.35/6.74	3.5496	8.2541
P12 + Base-Generated-SFT800/40	7.62	8.33	2.55/3.88	12.01/10.39	11.51/12.40	4.39/6.64	3.5411	8.2609
P12 + Selected-Original400/15	8.28	9.34	2.59/4.12	12.25/10.71	12.50/14.50	5.77/8.05	3.4746	8.2967
P12 + Random-Candidate400/15	7.62	8.53	2.54/3.98	12.20/10.67	11.33/12.59	4.41/6.89	3.5133	8.2388
P12 + Dense-Teacher-SFT400/15	7.68	8.57	2.61/4.14	7.16/5.32	15.53/16.83	5.41/7.99	3.5404	8.3762
P15 (Base)	8.91	9.73	3.00/4.40	12.17/10.88	13.54/15.07	6.94/8.56	3.4788	8.3662
P15 + Verifier-SFT400/15	8.00	8.59	2.78/4.02	12.27/10.78	12.39/12.48	4.54/7.08	3.5280	8.2871
P15 + Verifier-SFT800/40	7.68	8.73	2.72/4.24	12.09/10.81	11.09/12.44	4.83/7.43	3.5543	8.2912
P15 + Dense-Teacher-SFT400/15	8.82	9.44	2.93/4.55	7.74/5.81	18.72/18.65	5.90/8.76	3.5282	8.4419
P16 (Base)	9.87	10.80	3.65/5.30	12.69/11.09	15.58/17.13	7.57/9.69	3.4721	8.4053
P16 + Verifier-SFT400/15	8.82	9.50	3.38/4.54	12.35/11.02	14.02/14.07	5.54/8.35	3.5345	8.3495
P16 + Verifier-SFT800/40	8.32	9.27	3.21/4.78	12.36/11.12	12.09/13.20	5.62/7.96	3.5688	8.3476
P16 + Dense-Teacher-SFT400/15	10.16	10.80	3.78/5.28	8.01/6.16	21.75/22.44	7.08/9.31	3.5249	8.4836
Dense-24 Teacher	7.86	8.77	2.13/3.39	12.21/10.67	13.72/14.48	3.39/6.52	3.4724	8.3113

Macro ER rises from 8.36% at P_{13} to 8.75% at P_{15} and 11.31% at P_{16} , while UTMOS and SECS change little. These results motivate testing recovery at P_{15} and P_{16} . In the separate architecture benchmark, RTF falls from 0.633 for Dense-24 to 0.375 for P_{15} , a $1.69\times$ speedup. First-packet latency falls from 2.600 to 1.572 s. These efficiency checkpoints are distinct from the quality and recovery checkpoints. Peak allocated CUDA memory falls from 3382 to 2530 MiB. The P_{16} architecture has a slightly lower first-packet latency (1.535 s) but no further reduction in RTF (0.377). The measured quality and efficiency trends favor P_{15} among these configurations.

Relative to the corresponding Base checkpoints in Table 3, recovery with Verifier-SFT400/15 reduces Whisper/SenseVoice Macro ER by 0.62/0.83, 0.91/1.14, and 1.05/1.30 percentage points at P_{12} , P_{15} , and P_{16} , respectively. UTMOS increases and MCD decreases at each depth, while model parameter counts remain unchanged. SenseVoice is not used to select targets, so the improvement in its scores supports a gain in content accuracy beyond the Whisper screening metric. Of 24 per-language ASR error rates (three depths, four languages, and two recognizers), 23 decrease. The sole exception is Chinese Whisper CER at P_{15} (12.17% to 12.27%). The macro-average gains therefore span multiple languages.

SFT800/40 further improves both ASR measures at P_{12} and P_{16} . At P_{15} , Whisper Macro ER decreases from 8.00% to 7.68%, whereas SenseVoice Macro ER increases from 8.59% to 8.73%. Relative to their corresponding Base checkpoints, the P_{15} and P_{16} SFT800/40 models reduce Whisper/SenseVoice Macro ER by 1.23/1.00 and 1.55/1.53 percentage points, respectively. At P_{15} , UTMOS increases from 3.4788 to 3.5543 and MCD decreases from 8.3662 to 8.2912 alongside the reduction in Whisper Macro ER, although UTMOS is also used to rank accepted recovery candidates. The recovered P_{15} model has 635.50M total parameters versus 859.19M for Dense-24, with observed Whisper/SenseVoice Macro ERs of 7.68/8.73 versus 7.86/8.77. Its worst-group Whisper excess over Dense-24 remains 1.44 percentage points in Korean (4.83 versus 3.39). English WER is also 0.59 percentage points higher. Thus, the lower macro-average error does not imply uniform accuracy preservation across languages.

At P_{12} , Selected-Original yields slightly higher Macro ERs for Whisper and SenseVoice than Base (8.28/9.34 versus 8.15/9.23). Random-Candidate yields 7.62/8.53, and verified student recovery yields 7.53/8.40. Student-generated targets provide the larger observed gain, consistent with the on-policy rationale [17]. Differences in prompts prevent attributing the smaller verified-random gap solely to ASR screening. In separate training runs with 800 targets, combining Base and SFT400/15 targets yields 7.24/8.32 versus 7.62/8.33 for Base targets alone. At P_{15} , Dense-Teacher reduces Chinese Whisper CER from 12.17% to 7.74% but increases Japanese CER from 13.54% to 18.72%. Under the same 400-target/15-update setting, verified student targets reduce Whisper/SenseVoice Macro ER by 0.15/0.17, 0.82/0.85, and 1.34/1.30 percentage points relative to Dense-Teacher targets at P_{12} , P_{15} , and P_{16} , respectively. The gap in Japanese Whisper CER is especially large at P_{15} (12.39% versus 18.72%) and P_{16} (14.02% versus 21.75%), whereas the teacher targets reduce Chinese CER. Thus, the lower macro-average errors do not merely reflect improvements in a single language.

3. CONCLUSION

We presented CLWE-Prune, which ranks LLM blocks by worst-group recognition-error excess across four languages and recovers pruned models using ASR-screened student trajectories, with the acoustic decoder held fixed. At all four matched depths, the multilingual P_k configurations have lower Whisper Macro ER than EN-WER P_k ; at P_{12} , Japanese CER is 12.51% versus 16.25% for EN-WER P_{12} . With Verifier-SFT400/15, 23 of 24 per-language ASR error rates decrease relative to the corresponding Base checkpoints across P_{12} , P_{15} , and P_{16} . The P_{15} model recovered with SFT800/40 has 26.03% fewer parameters than Dense-24 and slightly lower observed Whisper/SenseVoice Macro ER. Relative to the corresponding Base checkpoints, Dense-Teacher targets reduce Chinese errors but increase Japanese errors at all three depths; under the matched 400/15 budget, verified student targets yield lower macro errors. Language-level gaps remain, notably in English and Korean. Future directions include joint selection and recovery, broader language coverage, and human evaluation.

4. COMPLIANCE WITH ETHICAL STANDARDS

This study used existing publicly available speech corpora and involved no new data collection or interaction with human participants. No additional ethical approval was required.

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